

Multiplication law



1 Simplify:

a $y^3 \times y^2$

b $x^6 \times 8x^3$

c $3y^6 \times 4y$

d $x^y x^z$

e $a^7 \times -6a^5$

f $-2a^4 \times 5a$

g $-b^7 \times -b^6$

h $a^b a^c$

i $5y^5 \times 6y^3$

j $-8u^3 \times -3u$

k $-4b^6 \times -5b^7$

l $x^a x^b x^c$

2 Complete the following statements:

a $b^4 \times b^{\square} = b^7$

b $7x^{15} \times \square = 35x^{27}$

c $c^{10} \times c^{\square} = c^8$

d $4x^{20} \times \square = 24x^{25}$

Division law

3 Simplify:

a $a^9 \div a^5$

b $\frac{x^{11}}{5x^8}$

c $\frac{6m^{15}}{m^4}$

d $b^6 \div b^2$

e $\frac{j^9}{j^2}$

f $x^7 \div x^4$

g $\frac{f^8}{f^5}$

h $s^{13} \div s^4$

i $\frac{4k^{11}}{2k^6}$

j $\frac{8g^{12}}{g^5}$

k $6p^{13} \div 2p^{10}$

l $15l^8 \div 3l^2$

4 Complete the following statements:

a $b^9 \div b^{\square} = b^5$

b $x^{\square} \div x^3 = x^2$

c $\frac{g^{\square}}{g^4} = g^7$

d $\frac{k^7}{k^{\square}} = k^3$



Power of a power law

5 Consider the expression $(p^3)^2$.

a State whether the following expressions are equivalent to $(p^3)^2$:

i $(p \times p \times p) \times (p \times p \times p)$

ii $p^3 \times p^3$

iii $(p \times p \times p)^2$

iv $p^3 \times p^2$

v $(p \times p \times p) \times (p \times p)$

b State whether the following equations are true or false:

i $(p^3)^2 = p^{3+2}$

ii $(p^3)^2 = p^{3 \times 2}$

c Complete the following:

$$(p^3)^2 = p^{\square}$$

6 Simplify:

a $(j^3)^6$

b $(w^2)^4$

c $(t^4)^3$

d $(3a^4)^3$

e $(5h^7)^3$

f $(xy^2)^5$

g $\left(\frac{2}{h^2}\right)^4$

h $\left(\frac{a^2}{b^5}\right)^3$

i $\left(\frac{3x^2}{y^5}\right)^3$

j $(-x^9)^4$

k $(-5x^4)^3$

l $\left(\frac{-2a^4}{3b^2}\right)^3$

7 Find the value of a and b in the following equation:

$$\frac{v^{18}}{w^{24}} = \left(\frac{v^a}{w^4}\right)^b$$

Zero index

8 Simplify:

a $18a^0$

b $(f^0)^9$

c $(g^{12})^0$

d $(6a)^0$

e $8r^0 + (2q^3)^0$

f $\left(\frac{2}{x^2}\right)^0$

g $\frac{3r^0}{t^2}$

h $8k^0 \div (4x^2)^0$

9 Complete the following statements:

$$\text{a } b^{11} \div b^{11} = b^{\square}$$

$$\text{b } \frac{h^{\square}}{h^8} = h^0$$



$$\text{c } \frac{v^9}{v^{\square}} = v^0$$

$$\text{d } y^{\square} \times y^5 = y^5$$

Mixed laws

10 Simplify:

$$\text{a } (x^6 y^3)^4$$

$$\text{c } p^5 \times (p^4)^3$$

$$\text{e } m^5 \div m^2 \times m^5$$

$$\text{g } (-2x^3)^4$$

$$\text{i } (7x^2)^0 - (8x^6)^0$$

$$\text{k } (4h^0)^2 + 18 \div (3g^0)$$

$$\text{m } (4u^5 v^2)^2$$

$$\text{o } (3y^5)^2 \times (5y^2)^3$$

$$\text{q } \frac{(2x^2 y^0)^4}{x^5}$$

$$\text{s } \frac{24a^3}{(2a)(4b)}$$

$$\text{u } \frac{3^{4a+2} \times 3^{1+6a}}{(3^3)^{3a-1}}$$

$$\text{b } \frac{y^7 \times y^6}{y^3 \times y^2}$$

$$\text{d } (2y^2)^3$$

$$\text{f } p^9 \div p^5 \div p^2$$

$$\text{h } \frac{6p^5 \times 4p^7}{8p^3}$$

$$\text{j } (12x^4)^0 + 12^0 - 12h^0$$

$$\text{l } \left((x^2)^6 \right)^5$$

$$\text{n } (3a^2 b^5 c)^4$$

$$\text{p } \frac{(x^4)^2}{x^5}$$

$$\text{r } (c^{10})^{11} \div (c^8)^3$$

$$\text{t } (10x^2 \times 10x^5)^0 - 10x^0$$

$$\text{v } \frac{81^{7a-4} \times 9^{3a+2}}{27^{3-3a}}$$

11 Write $(16^p)^4$ in the form a^b , where a is a prime number.